TRANSMISSION LINES & WAVEGUIDES – ECA1501

EXPERIMENT – 3

Designing a Quarter wave transformer by varying following Parameters a) frequency b) input impedance c) characteristic impedance

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AIM

To design a quarter-wave transformer for an input impedance (Zin) of 50 ohms and a load impedance (ZL) of 100 ohms and to analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz }.ZL =Zo.

SOFTWAREREQUIRED:

SCILABVersion6.1.1

PROCEDURE:

●Open SCILAB software

● Open file-new

● Put the input values

● Type the program

● Execute the program

● End the program

PROGRAM:

```
clear;clc;

Zr=100;s=9;d=0.1;

Zs=50;

Ro=sqrt(Zs*Zr);f=500*(10^6);r=d/2;

Zof=276*log10(s/r);
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Zoq=sqrt(Zr*Zof);

L=300/(f*(10^-6));

printf("length of lambda/4 transformer = %f meters\n",L);

printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);
```

TEST CASE 1

Design a quarter-wave transformer for an input impedance (Zin) of 50 ohms and a load impedance (ZL) of 100 ohms. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}.

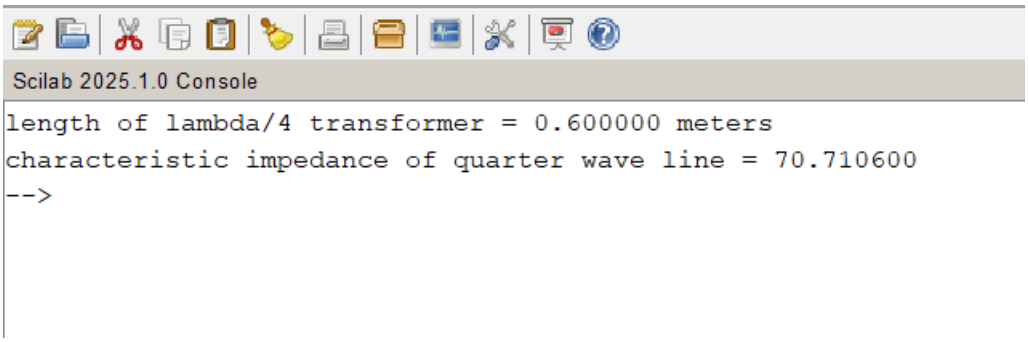
OUTPUT TABLE 1

ZR=100 ohms Zin =50 ohms

ZS ohms	ZR ohms	Ro ohms
50	100	70.71

ZR=100 ohms Zin =50 ohms

Frequency (MHz)	Length of λ/4 transformer (meters)
100	0.7500
200	0.3750
300	0.2500
500	0.1500



RESULT:

A quarter-wave transformer for an input impedance (Zin) of 50 ohms and a load impedance (ZL) of 100 ohms is designed and the transformer's performance analyzed for the varying operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz }.

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TEST CASE 2

To design a quarter-wave transformer a ftixed operating frequency of 2 GHz and a load impedance (ZL) of 100 ohms and evaluate its performance as the input impedance (Zin) varied .To validate that the transformer's characteristic impedance (Ztransformer= Ro= sqrt(Zin·ZL) changes with Zin.

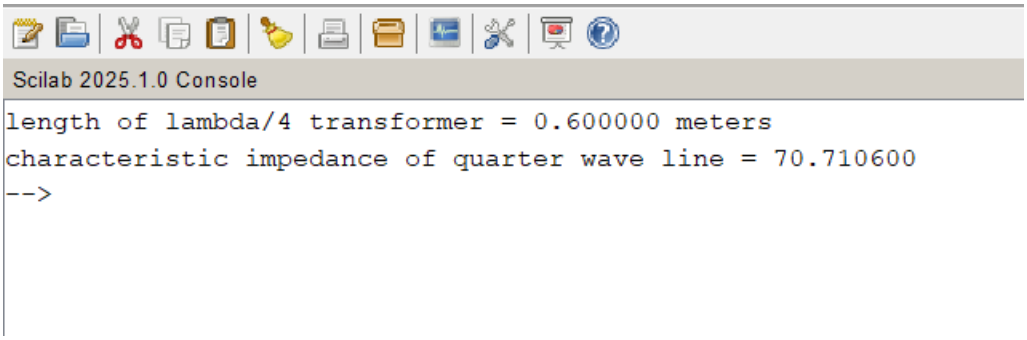
OUTPUT:

ZR=100 ohms Variable :Zin

Zin	Ro
25	50.00
50	70.71
75	86.60
100	100.00

Z_L=100,Omega), (f=2,GHz)

Characteristic Impedance \(R_o\) (Ω)	Input Impedance \(Z_{in}\) (Ω)
50	25
70.71	50
86.60	75
100	100



RESULT:

A quarter-wave transformer is designed at a fixed operating frequency of 2 GHz and load impedance (ZL) of 100 ohms and its performance is evaluated for various input impedance values. The variation of transformer’s characteristic impedance with Zin is also validated..